PES UNIVERSITY

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UE24MA241B – Linear Algebra and Its Applications

1 Linear Algebra Functions Reference - Unit 2 Notebook

A detailed reference for every linear algebra and matrix-related function used across all 5 tasks, including inputs, outputs, and internal mathematical operation.

1.1 1. scipy.linalg.orth

Used in: Task 2

```
col_space = orth(users_movies)
```

Details

Input	A - a 2D NumPy array (matrix) of shape (m, n)
Output	A 2D NumPy array of shape (m, r) where r = rank(A)

What it does mathematically: Uses SVD ($A = U\Sigma V^T$) to extract the r columns of U corresponding to non-zero singular values, forming an orthonormal basis for the column space C(A). Here it identifies the independent rating directions in the users-movies matrix.

1.2 2. scipy.linalg.null_space

Used in: Task 2

```
ns = null_space(users_movies)
```

Details

Input	A - a 2D NumPy array of shape (m, n)
Output	A 2D NumPy array of shape (n, k) where k = nullity(A) = n - rank(A)

What it does mathematically: Also uses SVD, returning the columns of V whose singular values are zero - these span N(A) = {x | Ax = 0}. Its dimension satisfies rank(A) + nullity(A) = n (Rank-Nullity Theorem), revealing redundancies in the users-movies matrix.

1.3 3. numpy.linalg.norm

Used in: Task 3

```
distance = np.linalg.norm(ratings[i, :] - trial_user.flatten())
```

Details

Input	x - a NumPy array (vector or matrix); optional ord parameter specifies the norm type (default is L2)
Output	A scalar float - the norm value

What it does mathematically: Computes the L2 norm $||u - v||_2$ = sqrt(sum of squared differences) of the difference vector between two rating rows. This is the Euclidean distance in R^{20}, used here to find which existing user rated the 20 movies most similarly to the trial user.

1.4 4. scipy.interpolate.interp1d

Used in: Task 4

```
f = interp1d(known_idx, known_ratings, kind='linear',
fill_value="extrapolate ")
```

Details

Input	x - 1D array of known x-positions (must be sorted); y - 1D array of known y-values; kind - interpolation method; fill_value - behaviour outside range
Output	A callable function object f such that f(x_new) returns interpolated values at new positions

What it does mathematically: Fills in a user's missing movie ratings by drawing straight lines between their known ratings, then reading off estimated values at every movie index.

1.5 5. cv2.getRotationMatrix2D

Used in: Task 5

```
rotation_affine = cv2.getRotationMatrix2D(center, angle, 1)
```

Details

Input	center - (cx, cy) pivot point in pixels; angle - rotation in degrees (positive = clockwise in OpenCV); scale - uniform scale factor
Output	A 2x3 NumPy float32 array - the affine rotation matrix

What it does mathematically: Builds the 2x3 affine matrix encoding a translate-rotate-translate-back sequence, rotating the image 45 degrees about its centre. This matrix is then passed directly into cv2.warpAffine to apply the transformation.

1.6 6. cv2.warpAffine

Used in: Task 5

```
stretched = cv2.warpAffine(ImJPG, affine_stretch, (int(n*scale_x),
int(m*scal e_y)))

rotated = cv2.warpAffine(ImJPG, rotation_affine, (n, m))
```

Details

Input	src - source image array (H, W, C); M - 2x3 affine transformation matrix; dsize - output canvas size (width, height)
Output	A transformed image array of shape (height, width, C)

What it does mathematically: Applies any 2x3 affine matrix M to every pixel via inverse mapping - here used twice: once with a hand-built scaling matrix to stretch the image by 1.5x, and once with the rotation matrix from getRotationMatrix2D to rotate it 45 degrees.

1.7 7. cv2.flip

Used in: Task 5

```
reflected = cv2.flip(ImJPG, 1)
```

Details

Input	src - source image array; flipCode - integer controlling the flip axis
Output	A new image array of the same shape with pixels reflected

What it does mathematically: Mirrors the image horizontally (flipCode=1) by mapping each pixel's x-coordinate to its mirror position, reversing left and right.